

1060 nm interferometric near-infrared spectroscopy

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Received 7 February 2025; revised 28 February 2025; accepted 3 March 2025; posted 4 March 2025; published 28 March 2025

Recently, interferometric near-infrared spectroscopy (iNIRS) has emerged to measure diffuse light field fluctuations with time-of-flight (TOF) resolution. Yet, current iNIRS implementations suffer from low signal-to-noise ratio (SNR). Longer wavelengths, with lower photon energy, lower reduced scattering in biological tissues, and higher permissible exposures, have the potential to increase SNR. Here, we investigate iNIRS at 1060 nm. Across various forehead locations, we find that the autocorrelation SNR is improved 3.7–9.3 times compared to 855 nm and 6.0–33.5 times compared to 773 nm at TOFs of 800–1000 ps. Physical system parameters account for much of this improvement, but the tissue response may also play a role. We conclude that wavelengths near 1060 nm can potentially improve iNIRS measurements of TOF-resolved speckle fluctuations. ©

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<https://doi.org/10.1364/OL.558899>

Diffuse optics, among the earliest and most mature approaches in the field of biophotonics, probes centimeters deep into biological tissues with multiple scattered near-infrared (NIR) light [1]. In diffuse correlation spectroscopy (DCS), NIR light speckle fluctuations encode deep blood flow from tissues such as the brain. For decades, investigators have used photon counting to measure this fluctuation signal [2,3]. Recently, DCS has been bolstered by developments such as photon counting arrays [4], longer wavelengths around 1060 nm [5,6], and deep learning [3]. While DCS is powerful, it faces challenges in quantitatively deciphering brain dynamics noninvasively through the intact scalp and skull in adults [7,8]. These challenges remain even at so-called “large” source-collector (S-C) separations of 4 cm where scalp contributions to autocorrelation decay rate can approach 45% [7].

Adding a time-of-flight (TOF or τ_c) dimension to DCS can help deal with superficial tissues quantitatively. Over the years, numerous techniques have emerged to provide TOF information about diffuse speckle fluctuations *in vivo* [9–14]. These approaches can be classified as either “direct” or “interferometric.” In the direct approach, known as time-domain DCS [10], a pulse of coherent light is injected into tissue, and detected photons are binned according to their TOF to provide

a temporal medium response. To achieve TOF resolution, the “direct” approach requires pulsed lasers and time-resolved photon counting. In the interferometric approach, a reference path is introduced, and the effective light temporal coherence provides TOF selection. These approaches include interferometric near-infrared spectroscopy (iNIRS) [12,15], path length resolved diffuse correlation spectroscopy [13], interferometric diffusing wave spectroscopy with a TOF filter [16], and dual comb approaches [14]. Critically, unlike the direct approach, the interferometric approach does not require fast detectors on the scale of the TOF resolution.

iNIRS provides simultaneous autocorrelations across TOFs at <100 ps resolution and, through digital shaping, achieves a nearly Gaussian impulse response function, making it unique among the above approaches. However, iNIRS signal has been limited by the collection of a single-mode fiber. Here, we aim to improve iNIRS by employing a longer wavelength of 1060 nm [5,6]. To characterize performance, we develop the autocorrelation signal-to-noise ratio (SNR) as a critical performance parameter. We characterize different TOF-resolved iNIRS noise regimes according to their dominant source of noise, namely, speckle or shot noise. We show that the longer wavelength clearly improves the autocorrelation SNR in the “photon-starved” shot noise-limited regime at late TOFs, which is critical to decipher brain blood flow index (BFI).

In our 1060 nm iNIRS setup [Fig. 1(a)], a distributed Bragg reflector (DBR) laser is coupled to a polarization-maintaining (PM) fiber by a module (DBR 1060PN from Thorlabs Inc.) with a specified internal isolation of 33 dB to attenuate backreflections and a polarization extinction ratio (PER) of 16 dB. To ensure proper attenuation of any unwanted (fast axis) polarization component, the light is passed through a polarizer with a specified PER of 26 dB and next to a PM coupler (PER: 20 dB) that splits the incoming light in a 90:10 ratio. The 90% arm is sent to the sample arm to illuminate the medium under investigation. The remainder, which is more than sufficient to put the system in the shot noise-limited regime, is directed to the reference arm and attenuated as needed.

The remitted light from the sample is combined with the reference light, which boosts the remitted sample light and serves as a TOF reference, by a 50:50 coupler (PER: 25 dB), with both outputs sent to a dual-balanced detector (DBD) (Newport 1817-FC). The entire fiber-based setup is constructed with a PM fiber

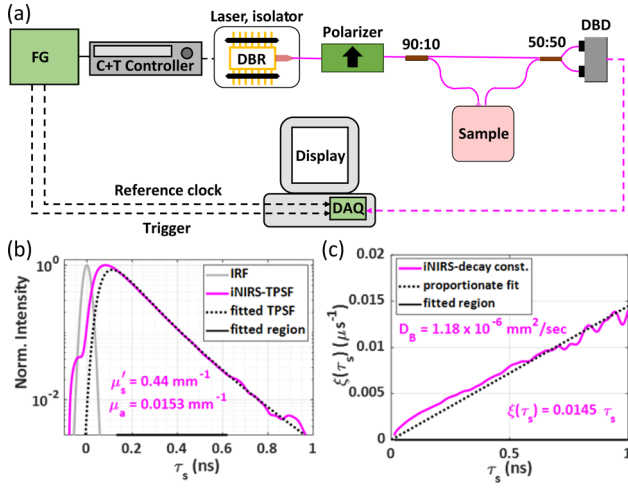


Fig. 1. 1060 nm iNIRS setup and validation. (a) Experimental setup (solid lines, optical paths; dashed lines, electrical paths; magenta, iNIRS path; black, electronic control signals). (b) Optical property recovery by fitting iNIRS TPSF and (c) dynamical property recovery by fitting $\xi(\tau_s)$ of Intralipid mixture. Abbreviations: FG, function generator; DBD, dual-balanced detector; 90/10, fiber coupler with 90:10 splitting ratio; 50/50, fiber coupler with 50:50 combining ratio; C + T, current and temperature; DAQ, data acquisition card.

and mating sleeves. Data acquisition and processing have been previously described [17].

The setup was used to measure the optical and dynamic properties of a mixture of 21 mL 20% Intralipid in 500 mL water with an expected μ'_s of 0.49 mm⁻¹ [18] and μ_a of 0.015 mm⁻¹ [19] at 1060 nm. The source-collector distance for the Intralipid measurement was set at 12 mm to enable the use of the diffusion theory for accurate optical property fitting. The recorded temporal point spread function (TPSF) was fitted to a diffusion model to obtain the mixture's optical properties [Fig. 1(b)]. Single exponential fitting of the TOF-resolved field autocorrelation yields a decay constant, $\xi(\tau_s)$, for every TOF, τ_s . The slope of $\xi(\tau_s)$ [Fig. 1(c)] along with the recovered μ'_s of 0.44 (95% CI: 0.43–0.46) mm⁻¹ were used to obtain the Brownian diffusion coefficient (D_B) of the Intralipid particles in water, via the method in [15]. The recovered D_B is 1.18×10^{-6} mm²/s (95% CI: 1.12–1.20 $\times 10^{-6}$ mm²/s).

We next compared performance with that of a dual-wavelength iNIRS system at 855 and 773 nm [17], via *in vivo* measurements on various locations on the forehead [Fig. 2(a)] of a healthy 28-year-old male human subject (Fitzpatrick type III skin tone) with a 40 s integration time. All human measurements were approved by the Institutional Review Board. Laser safety measures included ANSI-compliant power, secure fastening of an optical probe, continuous vigilance, and post-measurement skin checks. We commenced by collecting 1060 nm iNIRS measurements at each of four distinct forehead locations. After each measurement, we marked the measurement location with a biocompatible sticker. Then, at each location, we removed the sticker, cleaned the site, and then collected dual-wavelength iNIRS measurements [17] at the same location.

The power delivered to the tissue was 65 mW for 1060 nm, 38 mW for 855 nm, and 20 mW for 773 nm, corresponding to 2.1- and 4.5-times higher incident photon rate at 1060 nm

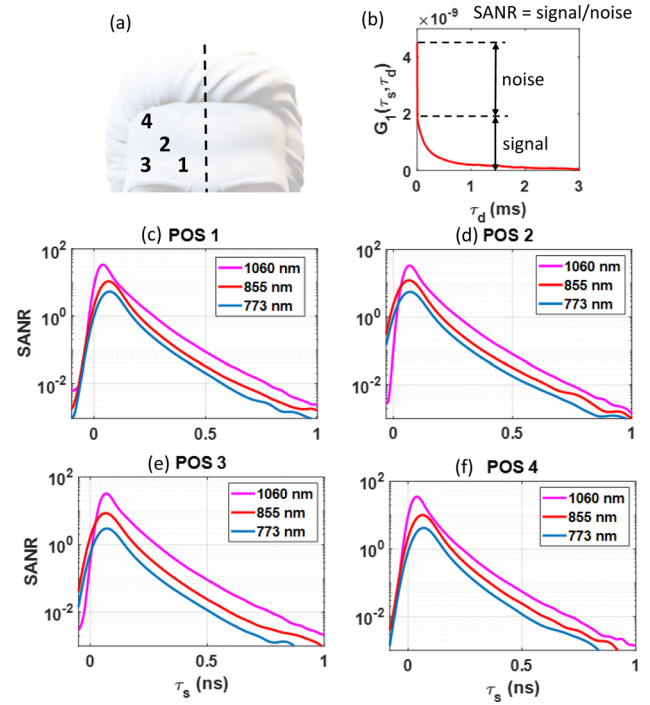


Fig. 2. *In vivo* comparison of iNIRS signal-to-additive noise ratio (SANR) across wavelengths on a human subject: (a) measurement positions 1–4; (b) SANR calculation; SANR versus τ_s at (c) position 1, (d) position 2, (e) position 3, and (f) position 4.

Table 1. iNIRS System Parameters at Three Studied Wavelengths^a

λ (nm)	P_{inc} (mW)	QE (%)	Digital Shaping Loss (dB)	$\Delta\tau_s$ (ps)	$h\nu$ ($\times 10^{-19}$ J)	System Merit (dB)
1060	65	82	2.28	40	1.87	0
855	38	65	2.30	76	2.32	-1.5
773	20	80	2.22	80	2.57	-3.5

^aAbbreviations: P_{inc} , incident power; QE, detector quantum efficiency; $\Delta\tau_s$, TOF resolution after digital spectral shaping [20]; $h\nu$, photon energy. The system merit factor, which determines the signal-to-additive noise ratio (SANR), is $10 \times \log_{10} (P_{\text{inc}} \times QE \times \Delta\tau_s / h\nu) - (\text{shaping loss})$ and defined as 0 dB for the 1060 nm system (first row).

compared to 855 nm and 773 nm, respectively. While the incident sample powers of the three systems were indeed different, these differences parallel the power of commercially available distributed feedback (DFB) and distributed Bragg reflector (DBR) laser sources, affording a practical comparison. The intensities were below the ANSI maximum permissible exposure for each wavelength. The tuning frequency and sweep rate were 100 kHz and 200 kHz, respectively. The power modulation was 42.8% for 1060 nm, 30.4% for 855 nm, and 69.4% for 773 nm. Other system parameters contributing to a system merit factor are shown in Table 1. For *in vivo* measurements, the S-C separation was 3.3 mm at all wavelengths.

Next, we briefly discuss two metrics for system comparison and their respective rationales. The first metric, the signal-to-additive noise ratio (SANR), is the ratio of the mean squared iNIRS heterodyne signal to the additive noise variance at each TOF (note that iNIRS should ideally operate in a regime where

shot noise from the reference arm dominates other additive noise sources [20]). The signal (numerator in SANR) is determined partly by the raw photon counts in the distribution of TOFs (DTOFs) at the collector fiber. First, digital spectral shaping reduces the SANR by introducing sub-optimal weighting of different frequencies [20]. Second, coarser TOF resolutions yield higher SANRs since they entail DTOF integration over a wider range of TOFs to form the TPSF. These and other relevant factors are summarized across wavelengths in Table 1. The second metric, the autocorrelation signal-to-noise ratio, SNR_{g_1} , is defined as the inverse of the variance of the normalized autocorrelation at lag k . If additive Gaussian noise dominates other noise sources, $\text{SNR}_{g_1}(\tau_s) = 2(N - k) \times \text{SANR}^2(\tau_s)$, where $k > 0$ and N is the total number of time points comprising the integration time. This simplifies to $\text{SNR}_{g_1}(\tau_s) \approx 2N \times \text{SANR}^2(\tau_s)$ for $N \gg k$. While SANR is a property of the raw iNIRS signal [Fig. 2(b)], as will be shown, SNR_{g_1} describes the field autocorrelation estimate, which is ultimately used to derive the blood flow index and is impacted by additional factors outside of SANR such as speckle.

We found that SANR was consistently higher at 1060 nm (improvement varied from 2.4 to 3.4 dB and from 4.6 to 6.6 dB across different locations for 855 and 773 nm, respectively, for τ_s from 0.8 to 1 ns) [Figs. 2(c)–2(f)]. The degree of improvement was partially explained by the system merit factor (Table 1).

We next asked whether the increased SANR at 1060 nm improved the SNR of the field autocorrelations, particularly at longer TOFs that are more sensitive to brain blood flow. The SNR of the iNIRS field autocorrelation, $\text{SNR}_{g_1}(\tau_s)$, is taken as the reciprocal of the variance of the estimate of $g_1(\tau_s, \tau_d)$ across the lag (τ_d) range from 8 to 10 ms, where the intrinsic sample field correlation has decayed completely leaving only noise. On average, for TOFs from 0.8 to 1 ns, 1060 nm improves SNR_{g_1} from 3.7 to 9.3 times over 855 nm and from 6.0 to 33.5 times over 773 nm (Fig. 3). Interestingly, at early TOFs, we observe different behaviors (Fig. 3), where SANR (from Fig. 2) has a sharp peak, highest at 1060 nm, but SNR_{g_1} (in Fig. 3) shows a broad plateau, and no wavelength is a clear winner. To investigate this apparent discrepancy, we compared SNR_{g_1} to SANR^2 (Fig. 4). We find that SNR_{g_1} is proportional to SANR^2 in the long TOF regime (>0.4 ns at 773 nm, >0.5 ns at 855 nm and >0.6 ns at 1060 nm), in agreement with the additive noise model. However,

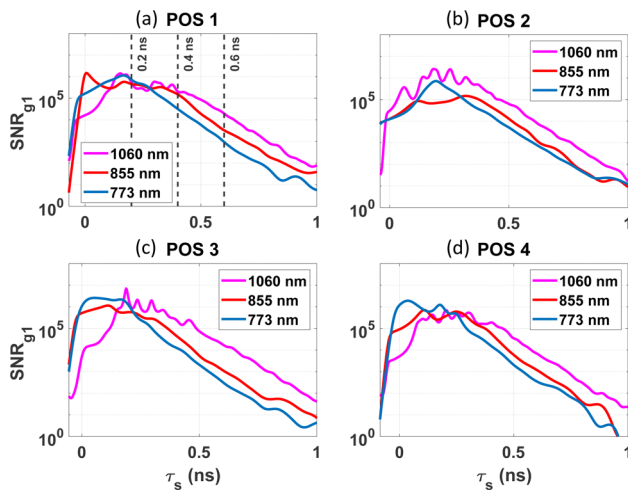


Fig. 3. Comparison of autocorrelation SNR (SNR_{g_1}) across wavelengths for positions (POS) 1 (a), 2 (b), 3 (c), and 4 (d), as shown in Fig. 2(a).

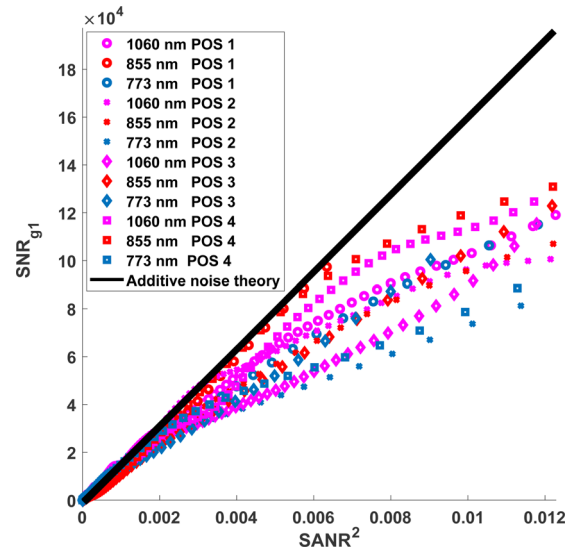


Fig. 4. Parametric plot of experimental SNR_{g_1} versus SANR^2 . The solid black line shows the theoretical relationship with just additive noise, which is approached in the shot noise limit at low SANR and late TOF. The deviation at high SANR and earlier TOF suggests the presence of another noise source.

for higher SANR at earlier TOFs, SNR_{g_1} is lower than predicted, suggesting the presence of an additional noise source.

The two identified noise regimes show distinct behavior [Figs. 5(a)–5(b)]. Specifically, the noise at an early TOF of 0.2 ns is mainly low frequency, while at the late TOF of 0.6 ns, the noise includes a broad range of frequencies, consistent with white noise (Fig. 5 insets). Supplement 1 contains a detailed analysis of the noise regimes. The white noise, prominent in the low SANR regime at 0.6 ns, is associated with shot noise, and its relative variance is inversely proportional to the number of detected sample photons squared [20]. Conversely, the low-frequency noise present in the high SANR regime at 0.2 ns, in both *in vivo* [Fig. 5(a)] and Intralipid measurements [Fig. 5(b)], is associated with speckle. Speckle noise variance is not related to the number of detected photons, but rather the number of speckle evolutions (i.e., decorrelation time).

We expect that our findings of speckle-limited autocorrelation SNR where photon counts are high should also apply in the field of DCS, where speckle noise is rarely discussed [21,22]. Note that unlike DCS, iNIRS can make autocorrelation measurements at early TOFs, with a slow decorrelation rate and a high sample photon count, two factors that contribute to speckle dominating other noise sources. Since speckle noise is invariant with the number of detected photons, increasing photon count in this regime is not expected to improve autocorrelation SNR, though increasing the number of speckles would. In contrast, improving photon counts at later TOFs (≥ 400 ps) benefits g_1 estimates, by improving the SANR. Importantly, late TOFs are more sensitive to brain blood flow if measured via the forehead [23,24]. Therefore, improving the signal at late TOFs, as achieved here by switching to 1060 nm, may aid an accurate estimation of brain blood flow.

Encouraged by these results, we performed 1060 nm iNIRS more comprehensively across nine different locations on a human forehead. The exponentially fit decay rates of measured autocorrelations, along with corresponding 95% confidence

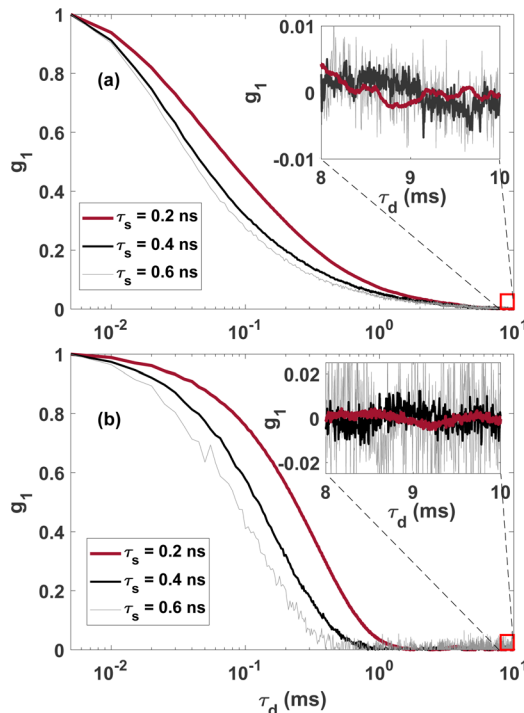


Fig. 5. (a) *In vivo* 1060 nm field autocorrelations (g_1) at three TOFs, labeled in Fig. 3(a). The inset highlights g_1 long after the decay, visualizing TOF-dependent noise characteristics. (b) Same as (a) but for Intralipid.

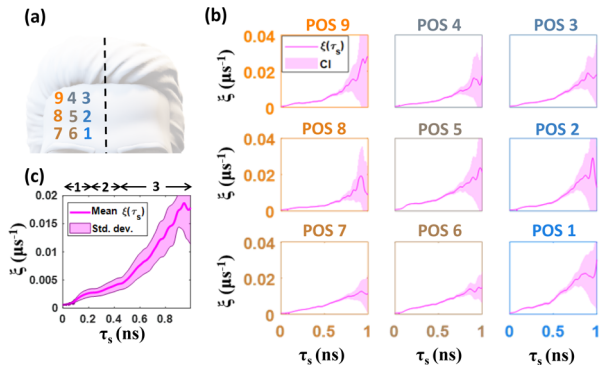


Fig. 6. 1060 nm iNIRS autocorrelation decay rates across the human forehead: (a) Measurement positions marked 1–9. (b) Decay rate, $\xi(\tau_s)$, from a single exponential fit and 95% confidence intervals across positions. (c) Grand average and standard deviation of $\xi(\tau_s)$ across positions with the three phases (1, scalp; 2, scalp and skull; 3, scalp, skull, and brain) demarcated.

intervals (Fig. 6), reveal triphasic behavior at most forehead locations and in the grand average [Fig. 6(c)]. This behavior, reported previously, arises due to differences in blood flow between the scalp (medium), skull (low), and brain (high), which are progressively sensed as TOF increases [23]. Though the triphasic behavior and the absolute values of $\xi(\tau_s)$ vary across locations due to differing local head geometries and perfusion states, a late phase, where the sensitivity to brain BFI causes the decay rate to increase rapidly [23], is consistently depicted (Fig. 6). By modifying layered dynamic models [23] to include TOF, from these data, it may be possible to retrieve layer thicknesses and blood flow indices.

This work highlights a late TOF regime in iNIRS, where shot noise determines autocorrelation SNR. The 1060 nm wavelength enabled a significantly higher autocorrelation SNR than conventional NIR wavelengths in this regime. The 1060 nm wavelength range also affords low loss fiber components, typically with better performance than in the conventional NIR range. Additionally, the higher optical diffusion coefficient (D) at 1060 nm allows deeper penetration and may provide brain sensitivity at earlier TOF than shorter wavelengths. These data support 1060 nm iNIRS for quantitative measurements of brain blood flow.

Funding. National Institutes of Health (EB029747, EB032840, EY031469, NS137832, EB035762); Research to Prevent Blindness (Stein Innovation Award, Unrestricted Grant).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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